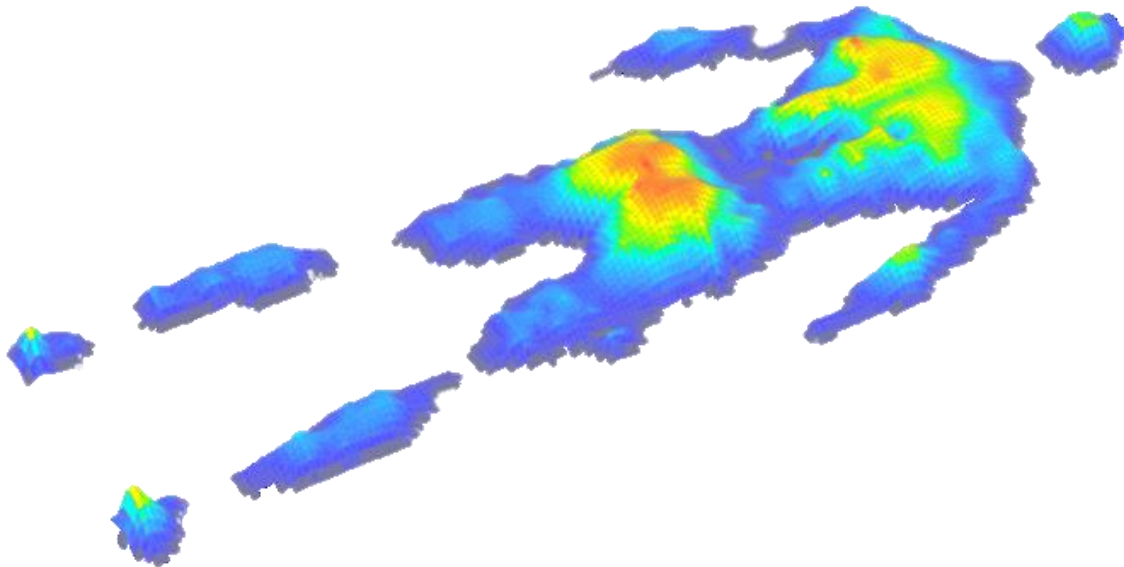


Mini-Study 1: Market Analysis and Technology Audit



Design of a Pressure Sensing Array for the Inflation Proning Repositioning Device

Harry Mills (12196)

Supervisor: Alexander Lunt

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Abbreviations

IPRD	Inflation Proning Repositioning Device
FSR	Force Sensitive Resistor
ARDS	Acute Respiratory Distress Syndrome
PP	Prone Position
PU	Pressure Ulcer
ICU	Intensive Care Unit
VS	Velostat
CNT	Carbon Nano Tube
CPI	Continuous Pressure Imaging

1 Introduction

This report covers initial research into the development of a pressure sensing array to be integrated with the Inflation Prone Repositioning Device (IPRD). The IPRD aids hospital staff during the repositioning of patients in the prone position (PP) who are suffering from ARDS. As outlined in the project brief[2], regularly repositioning patients is important to prevent pressure sores from forming, which can lead to longer hospital stays and further infection risk[3].

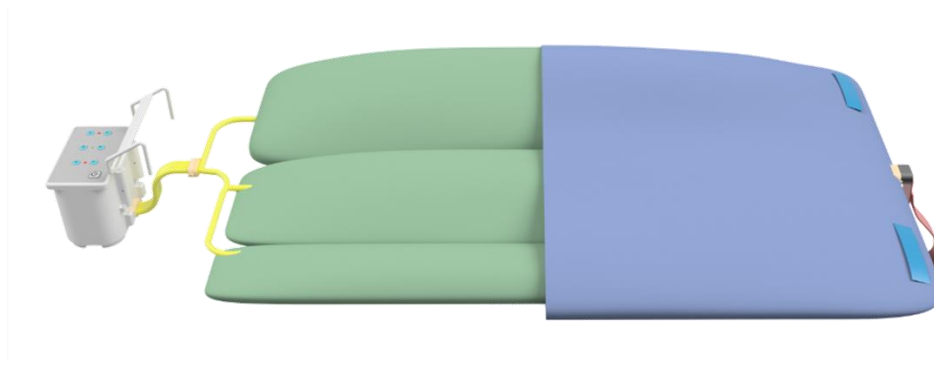


Figure 1 Current working model of the IPRD. This is placed below the patient's body and inflated to assist head and body movement during prone episodes[4]

The pressure sensing array to be developed in this project aims to detect pressure points between the patient and IPRD in real-time. Figure 2 shows how this will be integrated with the overall project. Initially this will be used for verification purposes to demonstrate the product will not cause pressure sores, but it could also be integrated into the final product to give information directly to hospital staff about where pressure sores could be forming.

This study covers the current technology available, commercial products, focussing on medically orientated products, and academic work in the space.

Initially, a technology audit of current pressure sensor arrays and their application in different industries and contexts will be done to develop an understanding of the capabilities and basic expectations of the system. The specific types of sensor architecture will then be compared within the context of integration into the IPRD, and finally conclusions from the analysis will be drawn to begin to development of a system requirements specification.

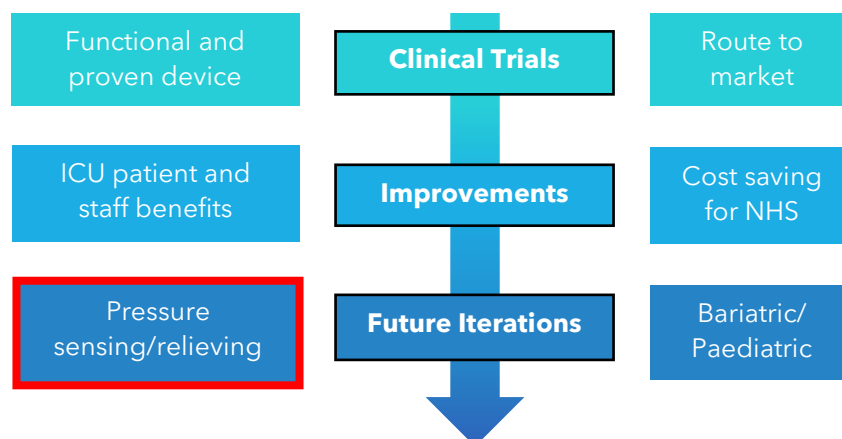


Figure 2 IPRD project roadma [4]. This project concerns pressure sensing/relieving (highlighted). This will not immediately be integrated into the commercial product but sooner will be used to verify the efficacy of the IPRD in not causing pressure points.

2 Technology Background

There are two primary ways of measuring the force applied to a point: load cells, and Force Sensitive Resistors (FSR).

Load cells are transducers which convert an applied load into a voltage via 4 strain gauges configured in a Wheatstone bridge[5]. Figure 3 shows a typical load cell.

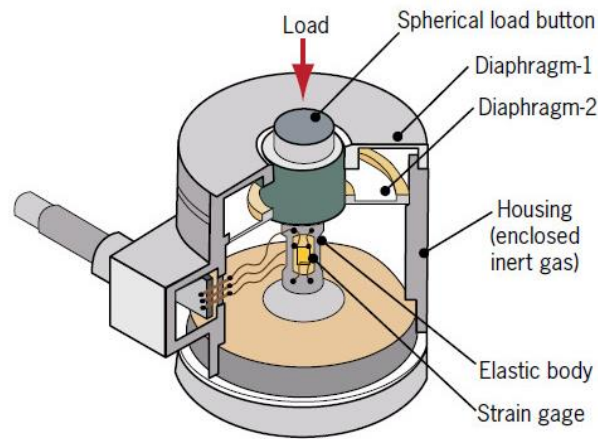


Figure 3 Cross-section of a button type load cell[6]

FSRs differ as they measure the force directly applied to the sensor rather than the substrate they are bonded to. They operate on the principle of piezoresistivity, meaning the electrical resistance changes when a force is applied. Figure 4 shows the construction of a typical FSR. A polymer film is used for the outer substrates, conductive pads are positioned either side of a piezoresistive layer, and an adhesive layer laminates the layers together [7].

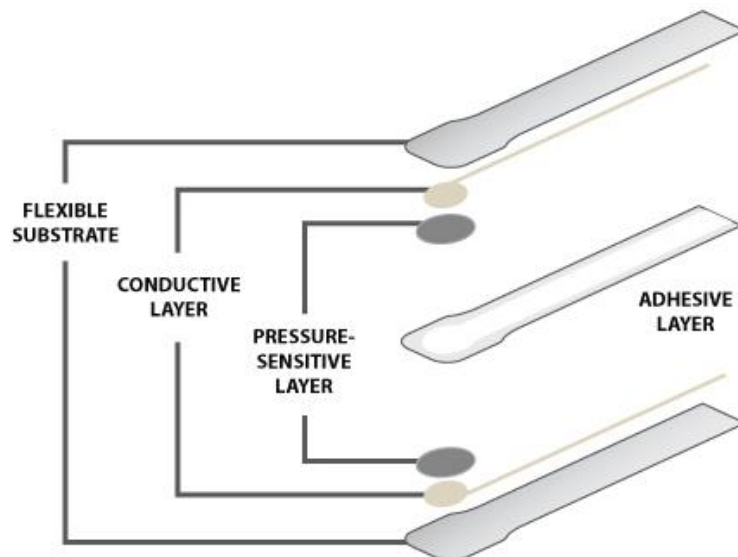


Figure 4 Construction of a typical FSR[7]

	Positives	Negatives
Load Cells 	Readily available Very accurate ($\pm 0.1\%$ [6]) Calibrated by manufacturer	Bulky and rigid Costly/complex electronics
Force Sensitive Resistors 	0.2 - 1mm thickness Flexible Low cost (Left: £7 [8]) Shock resistant Moisture resistant	Less accurate ($\pm 5\%$ [6]) Requires regular calibration Highly non-linear response

Table 1 Comparison of Load cells and FSR [6]

Table 1 compares FSRs and strain gauges. Strain gauges are far more accurate and work 'out of the box' with no need for calibration, but their bulky rigid construction means they are not suitable for this project as they would cause discomfort to the patient. FSRs are more suitable due to their thin construction and flexible, unobstructive nature. However, their low accuracy and need for careful calibration provides a challenge for implementation.

2.1 Matrix Pressure Sensors

The FSRs shown are single-point measurement devices so are not useful by themselves in this project. Pereira et al. (2020)[9], explores using an array of discrete FSRs to map body pressure on an airbed, but this approach provides insufficient resolution and becomes unwieldy when many sensors are used. (see Appendix A)

A more efficient method is to create a matrix of force sensitive cells (**sensels**). Pressure sensing arrays work by layering orthogonal conductive strips to create a matrix of contact points. This allows a densely packed grid to capture detailed pressure distributions over a surface. Figure 5 shows a force sensitive resistor array (also called a pressure mapping or tactile sensor).

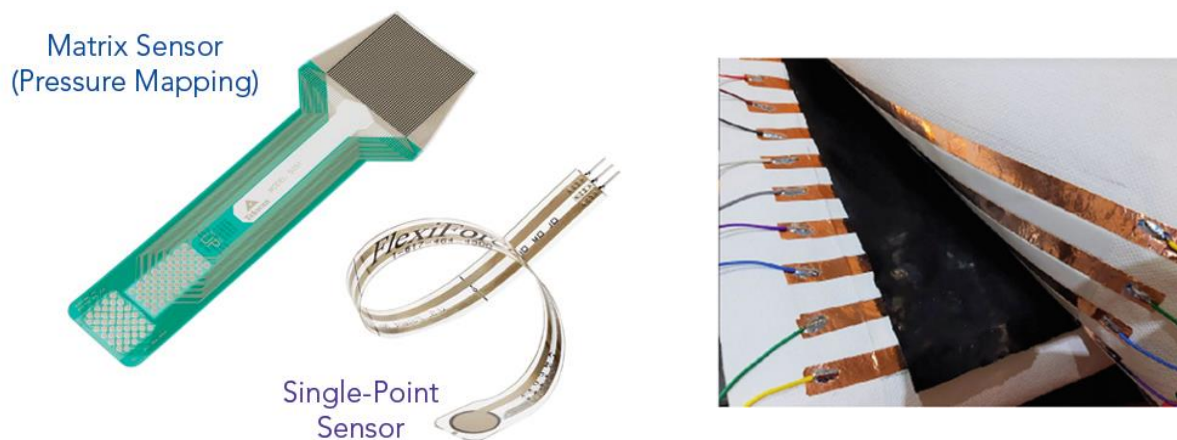


Figure 5 Left: Comparison of matrix pressure sensor and single-point FSR [7], Right: Construction of a matrix pressure sensor showing the perpendicular rows and columns, and piezoresistive layer.[11]

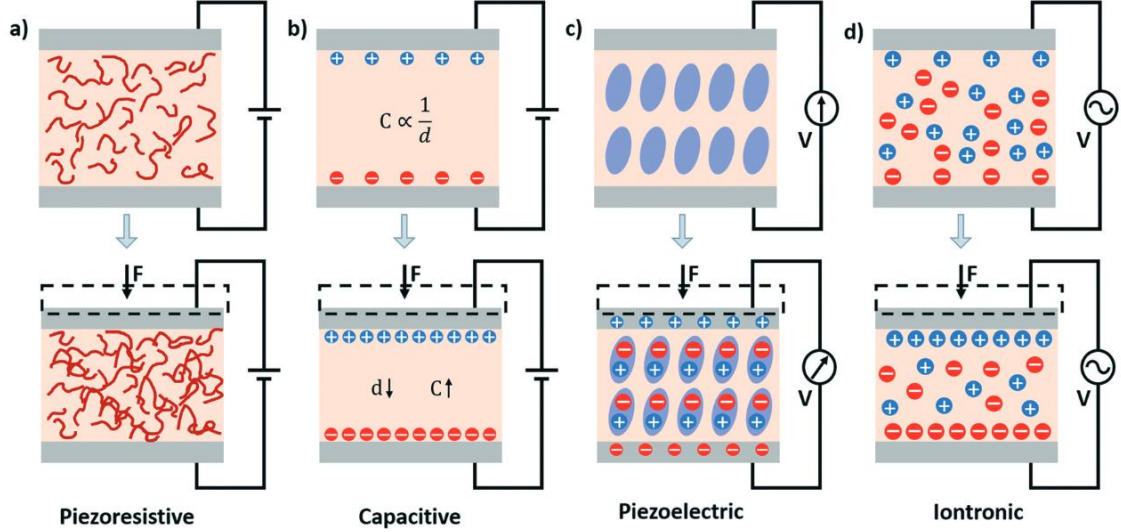


Figure 6 Schematics showing the main mechanisms of force transduction in pressure sensors[12]

2.1.1 Piezoresistive Matrix Pressure Sensors

Figure 5 shows a piezoresistive-based matrix pressure sensor. The basic principle is the same as the FSR (Figure 4), where an applied stress causes a change in the resistivity of the pressure-sensitive material, known as the piezoresistive effect.[13], [14]

Figure 6(a) shows the mechanism by which this change occurs. Due to Poisson's ratio (ν), materials that are elongated also contract in the transverse direction of elongation. The resistance (R) of a conductive material will change, according to:

$$R = \frac{\rho L}{A} \quad (1)$$

Where ρ is the resistivity, L is length, and A is the cross-sectional area of the conductor. Applying a compressive load decreases the resistance. Piezoresistivity is widely used in flexible sensors due to its high sensitivity and simple device designs.[12]

Piezoresistors which are silicon or metal-based (NiCr or Cu-Ni) exhibit strong piezoresistive responses but are susceptible to fracture when exposed to heavy loads on flexible substrates. Instead, polymer-based composites filled with conductive particles, like Carbon NanoTubes (CNTs), are favoured due to their versatility. CNT-polymer composites are widely used as mechanical strain sensors because they undergo resistance alterations in response to applied pressures and are very flexible.[12][15][16]

Currently available flexible piezoresistive strain sensors include **Velostat**(VS[17]), a flexible conductive polymer impregnated with carbon black, and conductive rubbers from Adafruit[18]. However, these products have limited stretchability (up to 70% strain), and low strain sensitivities (Gauge Factor = ~ 1)[12][19]. Velostat performance is also affected by temperature changes and susceptible to viscoelastic creep. As polymer chain entanglement is irreversible, the relaxation time of the material increases when compressed. When a constant force is applied the output can drift (i.e.: resistivity decreases with time). Drift, decreasing resistivity and response time, can compromise sensor accuracy and sensitivity, especially during prolonged use or fast-motion tasks[20]. Figure 7 shows the basic construction of a piezoresistive-based pressure sensor using Velostat to demonstrate how individual force sensing nodes are created at each intersection.

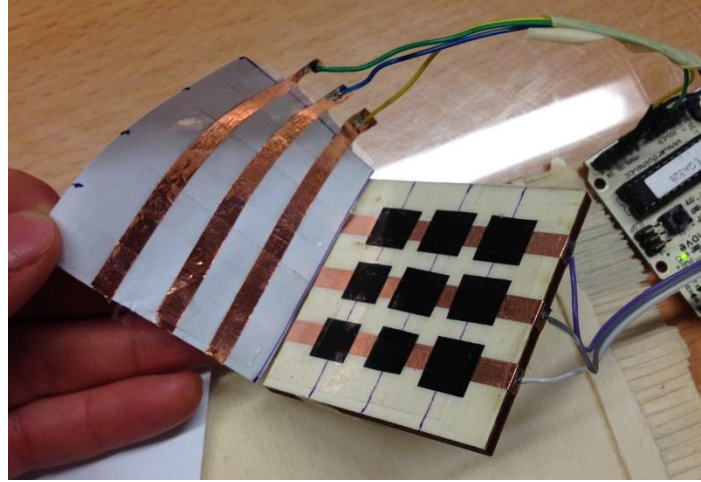


Figure 7 basic 3x3 piezoresistive based matrix pressure sensor, constructed by sandwiching Velostat pads between the contact points of the columns and rows[21]

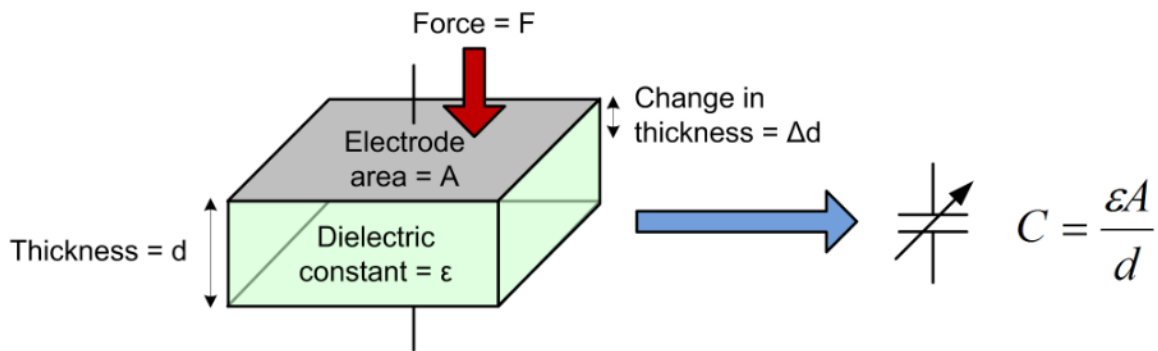


Figure 8 Detailed schematic of operating principle of capacitive pressure sensing[22]

2.1.2 Capacitive Matrix Pressure Sensors

Figure 8 shows the operating principle of capacitive tactile pressure sensors. These work by measuring the variations in capacitance in a parallel plate capacitor from an applied load. Capacitance (C) is related to the separation (d) and area (A) of the parallel plate capacitor, which uses an elastomeric dielectric (ϵ) to provide compliance [23]. The dielectric acts like a spring and compresses in response to applied force. Using the equation in Figure 8, capacitance changes as the gap decreases.

Processing these variations requires more complex electronics than piezoresistive arrays, and capacitive arrays tend to have higher hysteresis than piezo-based sensors.[22], [23].

To form an array, the dielectric is sandwiched between rows of electrodes on one side and columns of electrodes on the other side. Figure 9 shows the construction of a capacitive tactile array.

Capacitive arrays benefits are a higher resolution, higher sensitivity, long-term drift stability, low temperature sensitivity, low power consumption and sensing of normal or tangential forces, compared to piezoresistive-based sensors.[23]-[25]

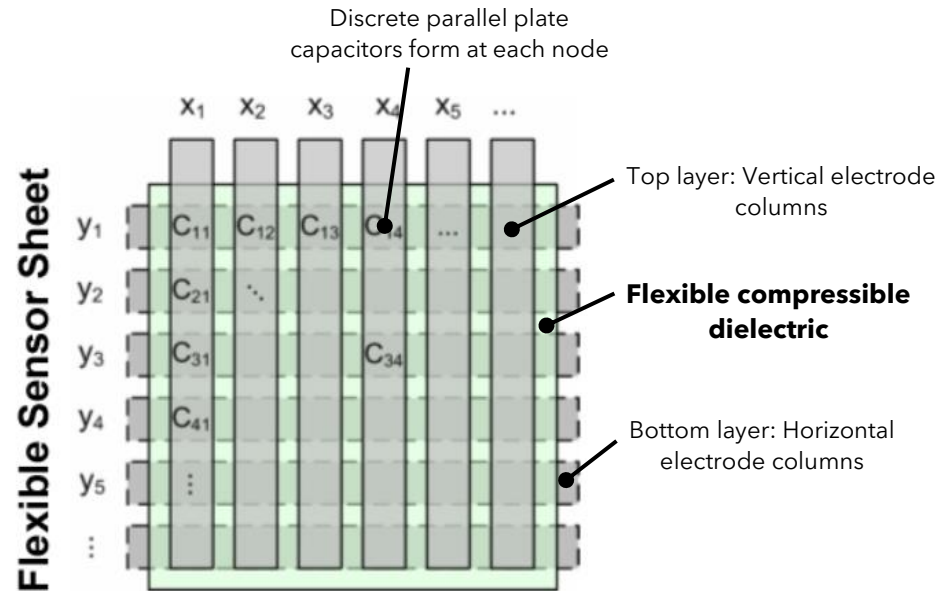


Figure 9 Basic construction of a capacitive-based tactile sensor array[22]

Selecting the elastic dielectric material considers: flexibility, capacitive range, repeatability and cost. A high permittivity dielectric (such as PVDF [14]) is also preferred as this maximises the change in capacitance over a range of pressures. Other materials were testing in Yip et al. (2009)[22], concluding that a flexible nylon textile was most suitable.

2.1.3 Alternative Tactile Architectures

The most common types of tactile pressure sensor construction are piezoresistive and capacitive (Figure 6). These have been found to be the most suitable for this project, however there are other methods explored below which did not meet the basic requirements.

2.1.3.1 Optical Sensors

Optical sensors measure mechanical contact, pressure, or movement using changes in light intensity or refractive index which can be detected using photodiodes. They can have very high spatial density (resolution) and have low signal-noise. However, incorporating light emitters and detectors (e.g.: CCD arrays [26]) increases bulk and rigidity, making them unsuitable for this project. [13], [23]

2.1.3.2 Piezoelectric Sensors

Certain crystals and some ceramics are piezoelectric; they generate a voltage potential when they are deformed[27]. The geometry of the element allows it to distinguish between transverse, longitudinal and shear forces, and the generated voltage is directly proportional to applied force, pressure or strain. Piezoelectric sensors have a very good high-frequency response, making them ideal for vibration measurement, but they are unable to measure static forces as the charge developed decays with time[13].

Sensor Architecture	Measured Parameter	Advantages	Disadvantages
Piezoresistive	Change in resistance	High spatial resolution High matrix scan rate Flexible & unobstructive Durable Low cost	Susceptible to drift Poor response to high frequency vibrations Low repeatability
Capacitive	Change in capacitance	Excellent sensitivity Good spatial resolution Large dynamic range Low power consumption Flexible & unobstructive Long-term drift stability	Noise susceptible Poor hysteresis effects
Optical	Light intensity	Fast scanning system response Very high spatial resolution Lightweight High repeatability Immune from EMI	Must include rigid and bulky light emitters and detectors
Piezoelectric	Change in voltage	Widely used for vibration sensing High frequency response High sensitivity	Limited to dynamic measurements Susceptible to temperature changes

Table 2 Comparison of various sensor architectures in the context of this project [12], [13], [23]

2.2 Applications of Tactile Pressure Sensors

Pressure mapping sensors are used in a wide range of industries: automotive safety testing, music, consumer product design and more. They are a highly effective means to measure interface pressure distributions, and due to their thin, unobtrusive construction are widely used in testing and research applications[28]. Medical uses are explored in detail in Section 3.1.

Figure 10-Figure 13 show some typical applications. Figure 10 shows capacitive sensors used for headphone pressure mapping. Capacitive sensors are more suitable for this as they have greater sensitivity than piezoresistive – allowing accurate measurement of the low pressure involved in the application[29]. Similarly, the applications shown Figures Figure 11-Figure 13 use the attributes described in Table 2 based on the context. For instance, piezoresistive sensors have a higher scan rate and tend to have higher resolution; so are more suited to fast changing detailed applications like Figure 11. Whereas capacitive sensors excel at measuring accurate pressures across a wide range, as in Figure 10 and Figure 12.

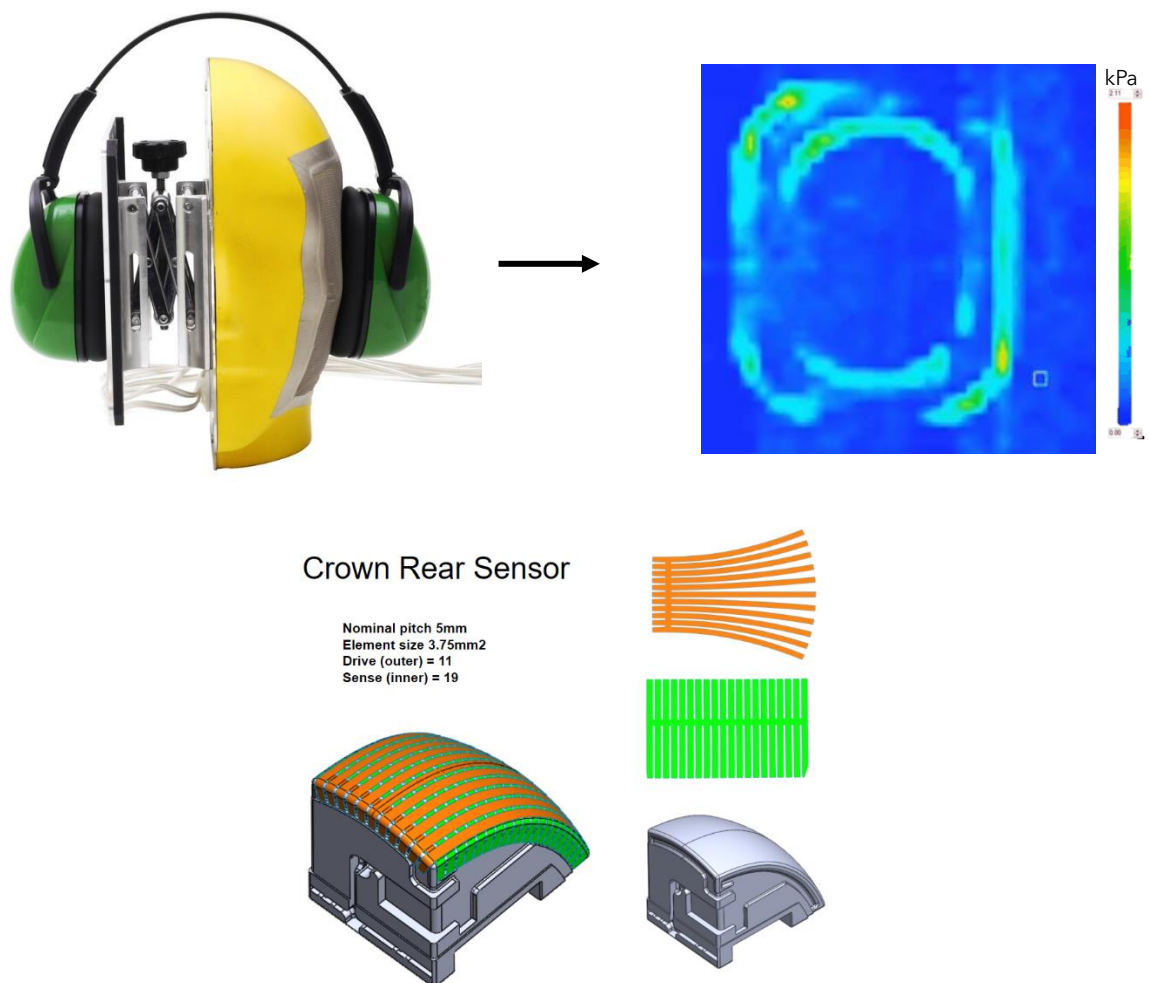


Figure 10 Head-shaped capacitive tactile sensors for testing consumer products[30]



Figure 11 Foot shaped piezoresistive tactile arrays for gait and posture analysis[31]

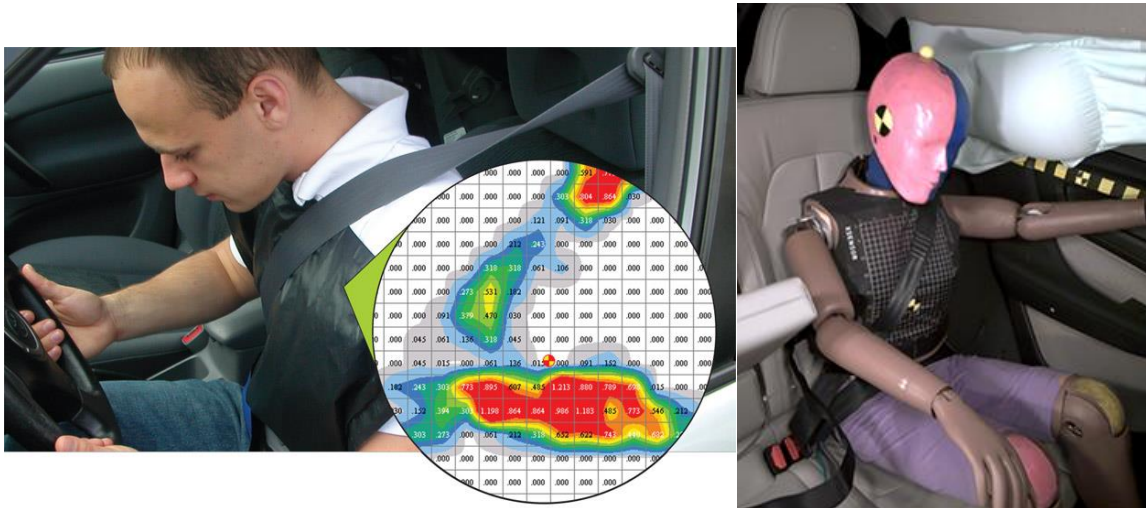


Figure 12 Piezoresistive (left) and Capacitive (right) Pressure sensing sheets to measure seatbelt tension during crash testing



Figure 13 Piezoresistive pressure sensitive array to measure bite pressure distribution in dentistry[32]

3 Medical Research

In the UK, the cost of treating pressure ulcers (PU) ranges from £1064 for an uncomplicated category I ulcer, up to £10500 per patient for serious category IV ulcers (see Appendix B: Pressure Ulcer Categorisation for PU categorisation scale). Annually, the cost of PU care in the NHS is estimated at between **£1.4 - £2.1 billion**. [33]

On average, patients who develop PUs can have their hospital stay extended by 4 days [34], and in over 75s it can be as long as 10 days [35]. Pressure ulceration in elderly patients is associated with a five-times increase in mortality, and in-hospital mortality in this group is 25%-33% [36].

3.1 Tactile Pressure Sensors for PU Prevention

As stated in the brief, the main aim of this project is to integrate pressure sensing capabilities into the IPRD to confirm it does not cause PUs to form, but also to feedback information to healthcare staff about potential pressure points. Using pressure sensing array to mitigate and prevent PU formation is not a new concept and has been studied in literature [11], [22], [37]-[41] and there are several commercial offerings which tackle the problem as well [42]-[46].

3.1.1 Commercial solutions

Currently, several pressure sensor arrays are offered in the market as full-body pressure mapping solutions (e.g.: Tekscan, Xsensor and Sensor Products Inc.).

Figure 14 shows the XSENSOR Foresite PT system. This is a thin mattress cover with an embedded array of (capacitive) pressure sensors providing **Continuous Pressure Imaging** (CPI) capability. This shows anatomical pressure points to doctors, patients and their families on a bedside LCD monitor. Healthcare staff can then utilise this data to determine when and how a patient should be repositioned to optimally redistribute pressure in real time [47], [48].

Behrent et al., (2014) [49] demonstrated that CPI can decrease the incidence of pressure ulcers in ICUs. In the study, only 0.9% of patients with CPI developed stage II PUs, compared to 4.8% of the control group. In another study done over a 2-month period, only one patient (0.03%) with continuous pressure mapping developed pressure ulcers, compared to 16 patients (5%) of the historical control group of patients placed on the same beds without CPI [50].



Figure 14 XSENSOR Foresite PT, pressure sensing hospital mattress cover [51]

A clinical trial with 678 participants, using the XSENSOR Foresite system was completed between 2014-2018 [48], [52]. This was done to evaluate the efficacy of CPI technology in the reduction of interface pressures and the incidence of PUs in ICUs. The study found “no statistically significant benefit” to using continuous pressure monitoring. However, the authors do suggest that a longer duration of monitoring and integrating CPI into a multifaceted intervention are required to see improvements in patient outcomes.

Similar body pressure sensing arrays are sold by Sensor Products Inc. [42], and Tekscan [43] (Figure 15). Specifications from these products has been combined in Table 3 to compare these products.

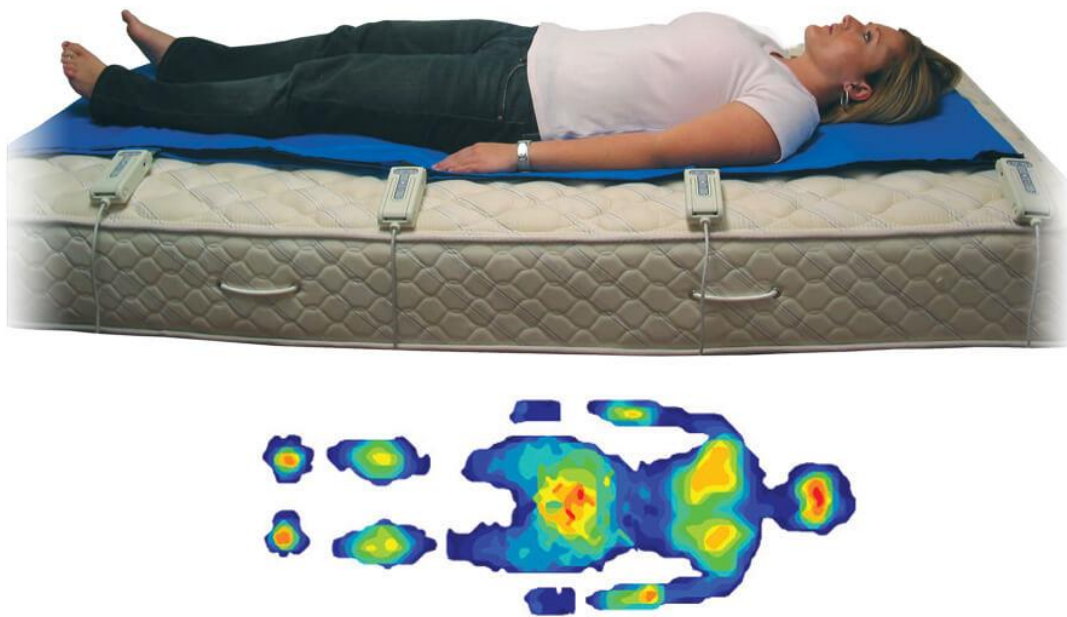


Figure 15 Tekscan BPMS [43]

	XSENSOR Foresite PT	Tekscan BPMS	SPI Tactilus Bodyfitter
Sensing technology	Capacitive	Piezoresistive	Piezoresistive
Cost	>\$10,000 ^[22]	>\$10,000 ^[22]	N/A
Scan frequency	1 Hz	60 Hz	50 Hz
Drift	5%/hr @ 100mmHg	5%/hr @ 100mmHg	-
Spatial resolution	15mm	10mm	25mm
Sensor range (mmHg)	5 - 200	0 - 300	0 - 100
Hysteresis	8%	4.5%	5%
Size (mm)	1880 x 760	1950 x 850	1850 x 760

Table 3 Comparison of 3 commercially available pressure sensor arrays designed for measuring patient-bed interface pressure in a medical setting [42][43][51]

Table 3 shows the main specification features of some body mapping pressure sensors on the market. The cost of these systems is orders of magnitude higher than the anticipated cost of the IPRD. This highlights the requirement for both a cheap testing and verification solution, but also a custom integrated continuous pressure monitoring solution that can be more tailored to the IPRD.

3.1.2 Literature Review

There have been a few studies in recent years exploring the design of low-cost pressure sensor arrays specifically to detect/prevent pressure ulcers.

Yip (2010)[22], proposes a capacitive-based sensor with a nylon dielectric for preventing pressure ulcers. This design is flexible, inexpensive, and provides continuous pressure monitoring. Due to the simple design and inexpensive materials (<\$50/m²), the sensor is also designed to be disposable.

Chung et al. (2015) presents an "inexpensive fabric-based pressure sensor array using off-the-shelf components that can be used to quantify pressure-related risk in decubitus ulcer formation". They claim this uses \$1 of parts for a 100x100mm square, so present this further as a disposable solution in medical settings. Interestingly, the author claims that some nurses find the automated alarms used in commercial bed-sized pressure sensors give excessive false alarms due to their excessive size; arguing that smaller sensors placed in particular areas of interest are more effective.

Yousefi et al. (2011), suggests a more holistic approach similar to the intended outcome of this project. They suggest a 'smart' hospital bed for both monitoring and prevention of PUs, by not just giving pressure distribution information but also automatically readjusting the patient. They argue that commercial solutions like the Tekscan BPMS, and XSENSOR Foresite cannot have a big enough effect alone, and propose a mechanical profiling bed (Figure 16) with feedback control from continuous surface pressure mapping.

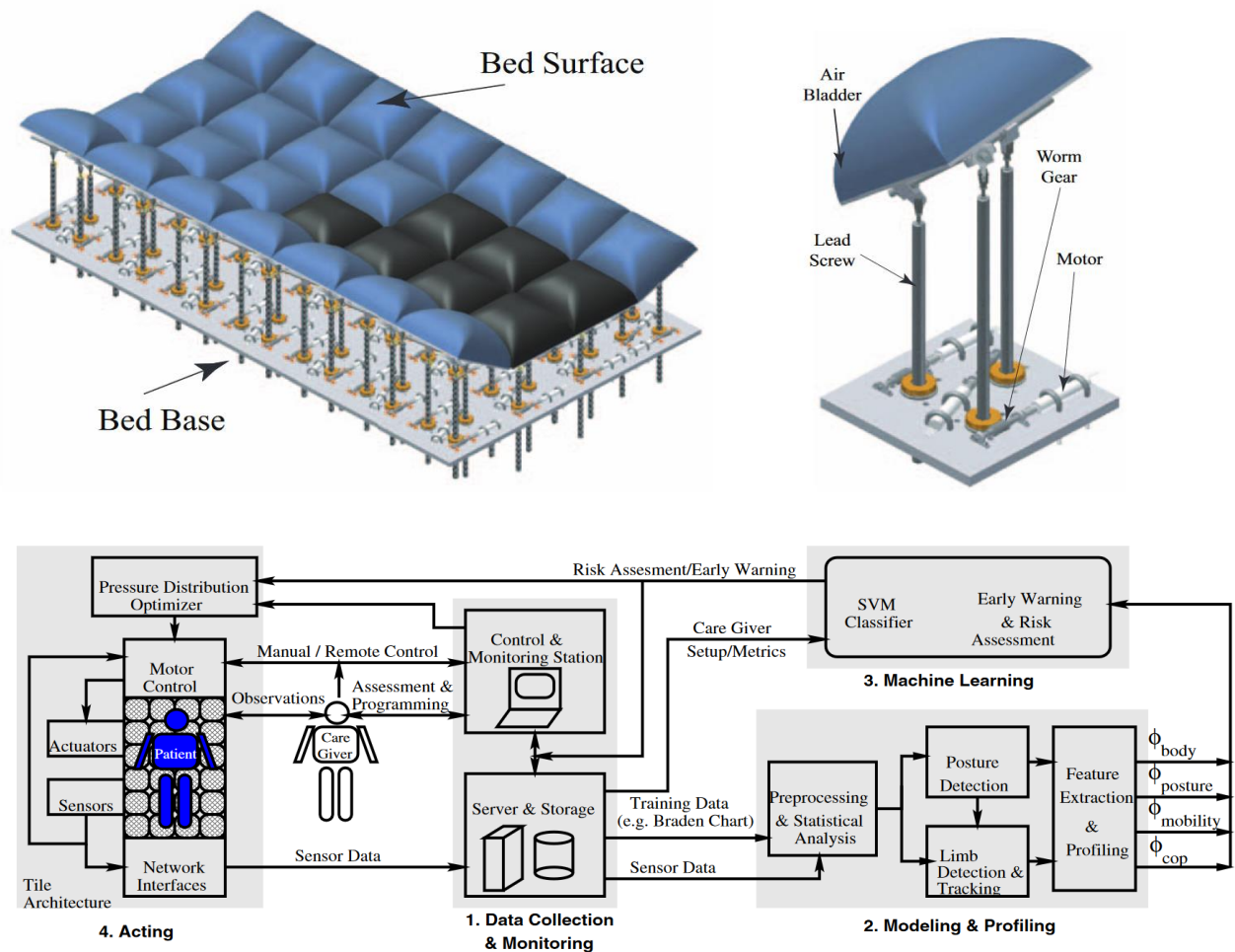


Figure 16 Top: Proposed mechanical repositioning bed from Yousefi et al. (2011). Bottom: High-level system architecture showing feedback loop to autonomously control bed position

Yousefi et al. is the most similar product or research I could find with respect to the project at hand. The pressure sensing method is different, and it incorporates a fully mechanical bed design, but the premise of using pressure distribution feedback to control patient repositioning (to prevent pressure ulceration) is an interesting topic not explored elsewhere that could be part of this project.

Yip (2010) and Chung (2015) are useful sources for design considerations of the pressure measuring sheet when considering specifically PU prevention. However, both studies stop short of developing the product into a medically usable prototype close to the level of commercial sensors (Table 3), so are more like a proof-of-concept of the sensor array/construction.

4 Conclusions

From the research conducted in this study a draft design specification can be drawn up with some initial high-level target requirements of the system (Table 4). These are based on analysis of available technology and comparisons of similar commercial products to highlight the fundamental design needs.

	Requirement	Demand/Wish	Source
1	Flexible, thin and unobtrusive construction	Demand	[2], [43]
2	Sterilisable container/outer surface	Wish	[47]
3	Fits within an NHS hospital bed (900 x 2000mm)	Demand	[2]
4	Visual feedback of localised pressure points	Demand	[41], [43], [47]
5	Integrate sensor array into IPRD surface	Wish	[47]
6	Closed-loop feedback to control IPRD inflation from surface pressure distribution	Wish	[40]
7	Low cost (<£150)	Wish	[11], [22], [41]
8	0.5hz refresh rate	Demand	[43], [47]
9	0 – 100mmHG pressure range	Demand	[43], [47]
10	25mm sensel resolution	Wish	[42], [43]

Table 4 Draft design specification concluding mini-study 1

After researching the fundamental sensor constructions, looking at commercial products – especially medical pressure sensors, highlighted the gap in the market for low-cost pressure mapping solutions. In the context of this project, spending up to £10,000 is not feasible, so there is a need for a low-cost custom implemented pressure sensing solution to be integrated with the IPRD. The sensor will be capacitive or piezoresistive based (TBD based on prototyping and material selection research) and will aim to meet the minimum requirements given in Table 4.

The next steps in this project will be to analyse the closely related research papers and numerous hobby/DIY projects around flexible pressure sensors to work towards an initial prototype, as well as research into material selection.

Another future research path is a focus on user needs. Input from doctors, nurses and other stakeholders will help to guide the technical development of the product.

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Appendix A: Discrete FSR matrix array

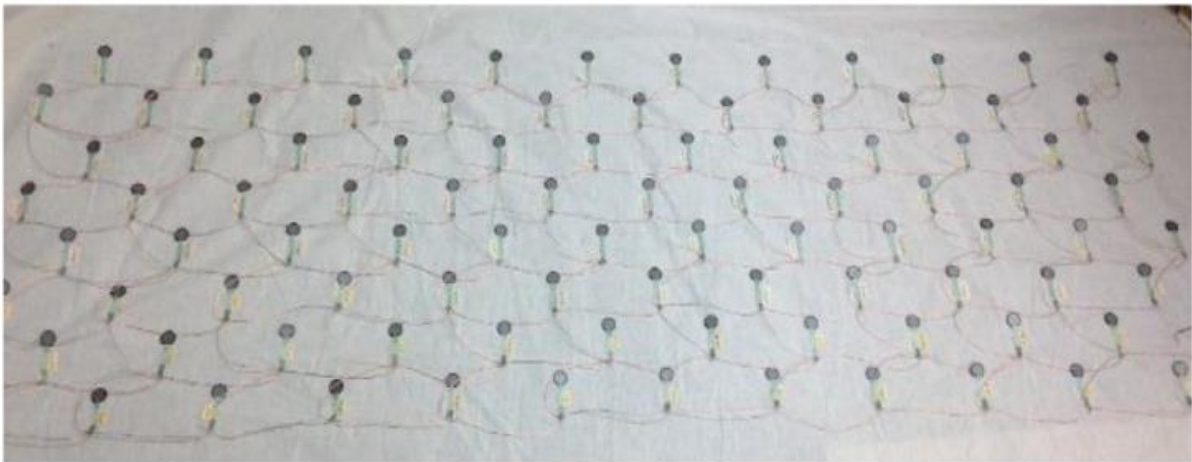


Figure 17 96 sensor FSR array [9]

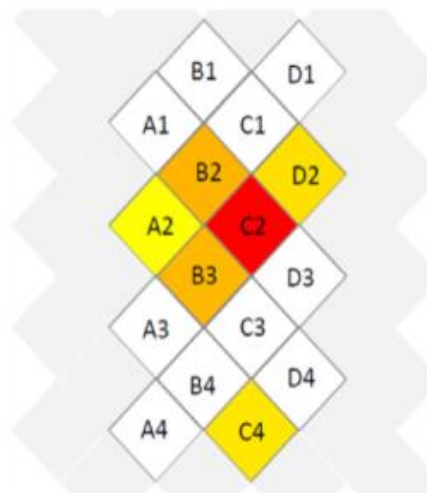


Figure 18 Air bed with user (left), achieved resolution (right) [9]

Discrete FSRs cannot provide good resolution and when a large area (e.g.: hospital bed) needs to be covered sufficiently this would require several hundred FSRs. Scaling this method will also make it much more expensive. Each FSR costs ~£5 so doubling the resolution or size will double the price. A more conventional integrated matrix array will not increase in price so rapidly when scaling – especially if the materials are bought in bulk.

Appendix B: Pressure Ulcer Categorisation

NHS guidance categorises pressure ulcers into 6 groups:

- Stage I
- Stage II
- Stage III
- Stage IV
- Suspected Deep Tissue Injury
- Unstageable: full thickness skin or tissue loss

More information (and graphic images) can be found in NHS pressure ulcer categorisation and guidance [53]